Essential Networking Concepts for SOC Analysts

Compiled By: Eman Elshimy

LinkedIn | GitHub

A revision guide for some SOC networking concepts. Covers OSI model, TCP/IP, ports, protocols, DNS, DHCP, ARP, ICMP, HTTP/HTTPS, firewalls, NAT, VLANs, VPNs, troubleshooting commands, packet analysis, and a glossary.

Table of Contents

1. OSI Model (7 Layers) 2
2. TCP/IP Model (4 Layers) 4
3. Ethernet and MAC Addresses 5
4. IP Addressing (IPv4 and IPv6) 6
5. Subnetting and CIDR 7
6. TCP 8
7. UDP 10
8. Common Ports and Protocols 10
9. DNS (Domain Name System) 12
10. DHCP (Dynamic Host Configuration Protocol) 13
11. ARP (Address Resolution Protocol) 15
12. ICMP (Internet Control Message Protocol) 16
13. HTTP and HTTPS 17
14. Firewalls and Network Security 19
15. NAT (Network Address Translation) 20
16. VLANs and Network Segmentation 20
17. VPNs (Virtual Private Networks) 21
18. Load Balancers and Proxies 22
19. Network Troubleshooting Commands 23
20. Packet Capture and Analysis 24
21. Quick Reference Tables (Keywords and Definitions) 26

1. OSI Model (7 Layers)

The OSI (Open Systems Interconnection) model is a conceptual framework that standardises network communication into seven layers. Each layer has a specific function and communicates only with the layers directly above and below it.

Layer 7: Application

Attribute	Description
Purpose	Provides network services directly to end-user applications
Protocols	HTTP, HTTPS, FTP, SMTP, DNS, SSH, Telnet, DHCP
Data unit	Data / Message
Devices	Gateways, firewalls (application-layer), proxies
Function	User interface, application-level communication

This is where users interact with the network. Most attacks target application-layer protocols; web attacks, email phishing, file transfers. Logs from web servers, email gateways, and authentication systems come from this layer.

Layer 6: Presentation

Attribute	Description
Purpose	Translates data between application and network formats
Functions	Encryption/decryption (SSL/TLS), compression, character encoding conversion
Examples	JPEG, GIF, MPEG, TLS, SSL, ASCII, EBCDIC
Data unit	Data / Message

TLS encryption happens at this layer. When you see HTTPS traffic, the actual HTTP data is encrypted here. This is why you cannot see inside encrypted packets without decryption capabilities.

Layer 5: Session

Attribute	Description
Purpose	Manages sessions (connections) between applications
Functions	Session establishment, maintenance, termination, authentication, checkpointing
Protocols	NetBIOS, RPC, PPTP, SMB (session component)
Data unit	Data / Message

Session hijacking attacks target this layer. Attackers steal session tokens to impersonate authenticated users without needing passwords.

Also note: While TCP handles connection *establishment* (Layer 4), the Session layer manages checkpointing and recovery. In modern security, TLS session resumption and cookies/session tokens (often analysed at Layer 5/7 boundaries) are prime targets for hijacking.

Layer 4: Transport

Attribute	Description
Purpose	Provides reliable or unreliable data delivery between hosts
Protocols	TCP (reliable, connection-oriented), UDP (unreliable, connectionless)
Functions	Port numbering, segmentation, flow control, error checking
Data unit	Segment (TCP) or Datagram (UDP)
Devices	Firewalls (stateful), load balancers

This is where port numbers live. TCP vs UDP determines how data is sent. Port scanning detects open services at this layer. Stateful firewalls track TCP connection states.

Layer 3: Network

Attribute	Description
Purpose	Routes packets between networks
Protocols	IP (IPv4, IPv6), ICMP, ARP*, RIP, OSPF, BGP
Functions	Logical addressing (IP addresses), routing, fragmentation
Data unit	Packet
Devices	Routers, layer 3 switches, firewalls

IP addresses are assigned at this layer. Routing decisions happen here. ICMP (ping) is used for network discovery and attacks (ICMP tunneling). This is where network segmentation and firewall rules operate.

**ARP straddles L2/L3 but is critical for L2 attacks.*

Layer 2: Data Link

Attribute	Description
Purpose	Provides node-to-node data transfer within the same network
Protocols	Ethernet, Wi-Fi (802.11), PPP, ARP
Functions	MAC addresses, frame delimiting, error detection (CRC)
Data unit	Frame
Devices	Switches, bridges, network interface cards (NICs)

MAC addresses are assigned at this layer. ARP poisoning attacks target this layer. Switches operate at Layer 2 and do not understand IP addresses. VLANs segment networks at this layer.

Layer 1: Physical

Attribute	Description
Purpose	Transmits raw bits over physical medium
Media	Copper cables (Ethernet), fibre optic, wireless radio
Functions	Bit encoding/decoding, signal transmission/reception

Data unit	Bits
Devices	Hubs, repeaters, cables, connectors

Physical layer attacks require physical access. Cable taps, rogue devices plugged into network jacks. This is why physical security matters.

OSI Model Mnemonic

Direction	Mnemonic
Bottom to top	Please Do Not Throw Sausage Pizza Away
Top to bottom	All People Seem To Need Data Processing

Layer	Bottom-Up	Top-Down
7	Please	All
6	Do	People
5	Not	Seem
4	Throw	To
3	Sausage	Need
2	Pizza	Data
1	Away	Processing

2. TCP/IP Model (4 Layers)

The TCP/IP model is the practical implementation used on the internet. It has four layers that map roughly to the OSI model.

TCP/IP Layer	OSI Mapping	Protocols
Application	Layers 5,6,7	HTTP, DNS, SMTP, FTP, SSH
Transport	Layer 4	TCP, UDP
Internet	Layer 3	IP, ICMP
Network Access	Layers 1,2	Ethernet, Wi-Fi, ARP

Layer Comparison

OSI Model	TCP/IP Model	Example
Application	Application	HTTP
Presentation	Application	TLS
Session	Application	NetBIOS
Transport	Transport	TCP
Network	Internet	IP

Data Link	Network Access	Ethernet
Physical	Network Access	Cable

3. Ethernet and MAC Addresses

MAC Address (Media Access Control)

Attribute	Description
Length	48 bits (6 bytes)
Format	Hexadecimal, usually written as 00:1A:2B:3C:4D:5E
Uniqueness	Assigned by manufacturer. Intended to be unique, often spoofed.
Permanence	Hardcoded into network interface (can be spoofed)

MAC Address Structure

Component	Bits	Description
OUI (Organizationally Unique Identifier)	First 24 bits	Identifies manufacturer
Device Identifier	Last 24 bits	Unique to the device

Special MAC Addresses

Address	Purpose
FF:FF:FF:FF:FF:FF	Broadcast (all devices on local network)
01:00:5E:xx:xx:xx	IPv4 multicast
33:33:xx:xx:xx:xx	IPv6 multicast

Ethernet Frame Structure

Field	Size	Description
Preamble	7 bytes	Synchronisation
SFD	1 byte	Start frame delimiter
Destination MAC	6 bytes	Receiver
Source MAC	6 bytes	Sender
EtherType / Length	2 bytes	Protocol type or frame length
Payload	46-1500 bytes	Data (IP packet)
FCS	4 bytes	Frame check sequence (error detection)

4. IP Addressing (IPv4 and IPv6)

IPv4 Address

Attribute	Description
Length	32 bits (4 bytes)
Format	Dotted decimal: 192.168.1.1
Total addresses	4,294,967,296 (2^32)
Structure	Network portion + Host portion

IPv4 Address Classes (Historical — now uses CIDR)

Class	Leading Bits	Range	Subnet Mask	Purpose
A	0	1.0.0.0 - 126.255.255.255	255.0.0.0	Large networks
B	10	128.0.0.0 - 191.255.255.255	255.255.0.0	Medium networks
C	110	192.0.0.0 - 223.255.255.255	255.255.255.0	Small networks
D	1110	224.0.0.0 - 239.255.255.255	N/A	Multicast
E	1111	240.0.0.0 - 255.255.255.255	N/A	Reserved

Private IP Addresses (Not Routable on Internet)

Range	CIDR	Number of IPs	Common Use
10.0.0.0 - 10.255.255.255	10.0.0.0/8	16,777,216	Large enterprise networks
172.16.0.0 - 172.31.255.255	172.16.0.0/12	1,048,576	Medium networks
192.168.0.0 - 192.168.255.255	192.168.0.0/16	65,536	Home/SMB networks

Special IPv4 Addresses

Address	Purpose
0.0.0.0	Default route, any address
127.0.0.0/8	Loopback (localhost), usually 127.0.0.1
169.254.0.0/16	Link-local (APIPA), when DHCP fails
255.255.255.255	Limited broadcast
224.0.0.0 - 239.255.255.255	Multicast
192.0.2.0/24	Documentation and examples (TEST-NET)

IPv6 Address

Attribute	Description
Length	128 bits (16 bytes)
Format	Hexadecimal, colon-separated: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Total addresses	340 undecillion (2^128), practically infinite
Abbreviation rules	Leading zeros can be omitted; consecutive zero blocks replaced with ::

IPv6 Address Types

Type	Prefix	Description
Unicast	2000::/3	Globally routable
Link-local	FE80::/10	Local network only (like 169.254.x.x)
Unique local	FC00::/7	Private (like RFC 1918)
Multicast	FF00::/8	One-to-many
Loopback	::1	Localhost

5. Subnetting and CIDR

What is Subnetting

Subnetting divides a large network into smaller, manageable subnetworks. It improves security, reduces broadcast traffic, and optimises routing.

CIDR Notation (Classless Inter-Domain Routing)

CIDR notation combines an IP address with a subnet mask: 192.168.1.0/24

CIDR	Subnet Mask	Number of IPs	Usable IPs	Common Use
/32	255.255.255.255	1	1	Single host
/31	255.255.255.254	2	2	Point-to-point links
/30	255.255.255.252	4	2	Small WAN links
/29	255.255.255.248	8	6	Very small networks
/28	255.255.255.240	16	14	Small networks
/27	255.255.255.224	32	30	Small office
/26	255.255.255.192	64	62	Medium networks
/25	255.255.255.128	128	126	Medium networks
/24	255.255.255.0	256	254	Common LAN size
/23	255.255.254.0	512	510	Larger networks
/22	255.255.252.0	1024	1022	Larger networks
/21	255.255.248.0	2048	2046	Campus networks
/20	255.255.240.0	4096	4094	Large networks
/16	255.255.0.0	65,536	65,534	Large enterprise
/8	255.0.0.0	16,777,216	16,777,214	Major organisations

How to Calculate Subnets

Step	Action
1	Identify the network bits (1s in subnet mask)
2	Identify the host bits (0s in subnet mask)
3	Number of subnets = 2^(borrowed bits). In modern networks, all subnets are usable.
4	Number of hosts = 2^(host bits) - 2

Network, Broadcast, and First/Last Host

Component	Calculation
Network address	All host bits set to 0
First usable host	Network address + 1
Last usable host	Broadcast address - 1
Broadcast address	All host bits set to 1

6. TCP

TCP Header Structure

Field	Size	Description
Source Port	16 bits	Sending application port
Destination Port	16 bits	Receiving application port
Sequence Number	32 bits	Ordering of data
Acknowledgment Number	32 bits	Next expected byte
Data Offset	4 bits	Header length
Reserved	3 bits	Reserved for future use
Flags	9 bits	Control flags (SYN, ACK, etc.)
Window Size	16 bits	Flow control buffer space
Checksum	16 bits	Error detection
Urgent Pointer	16 bits	Urgent data location
Options	Variable	Additional parameters

TCP Flags (Important for SOC)

Flag	Name	Purpose
SYN	Synchronise	Initiates connection
ACK	Acknowledgment	Confirms receipt of data
RST	Reset	Abruptly terminates connection

FIN	Finish	Gracefully terminates connection
PSH	Push	Send data immediately
URG	Urgent	Urgent data

TCP Three-Way Handshake

Step	Direction	Flags	Purpose
1	Client → Server	SYN	Client requests connection
2	Server → Client	SYN-ACK	Server acknowledges and agrees
3	Client → Server	ACK	Client confirms, connection established

Detection relevance: SYN packets without ACK responses indicate port scanning.

TCP Four-Way Termination

Step	Direction	Flags	Purpose
1	Client → Server	FIN	Client requests to close
2	Server → Client	ACK	Server acknowledges
3	Server → Client	FIN	Server agrees to close
4	Client → Server	ACK	Client acknowledges, connection closed

TCP Connection States

State	Description
CLOSED	No connection
LISTEN	Waiting for connection
SYN-SENT	Sent SYN, waiting for SYN-ACK
SYN-RECEIVED	Received SYN, sent SYN-ACK
ESTABLISHED	Connection open, data flowing
FIN-WAIT-1	Sent FIN, waiting for ACK
FIN-WAIT-2	Received ACK, waiting for FIN
CLOSE-WAIT	Received FIN, sent ACK
LAST-ACK	Sent FIN, waiting for final ACK
TIME-WAIT	Waiting before closing completely

7. UDP

UDP Header Structure

Field	Size	Description
Source Port	16 bits	Sending application port
Destination Port	16 bits	Receiving application port
Length	16 bits	Header + data length
Checksum	16 bits	Error detection (optional)

UDP Characteristics

Characteristic	Description
Connectionless	No handshake, no connection state
Unreliable	No guarantee of delivery
No ordering	Packets may arrive out of order
No flow control	Sender can overwhelm receiver
No error recovery	Lost packets are not retransmitted
Lower overhead	Smaller header (8 bytes vs 20+ for TCP)

When UDP is Used

Application	Why UDP
DNS	Small queries, retry is simple
DHCP	Broadcast-based, no connection needed
SNMP	Network monitoring, occasional loss acceptable
VoIP / Video streaming	Speed > reliability
Gaming	Real-time, lost packets discarded
Syslog	Simple logging, UDP is faster

8. Common Ports and Protocols

Critical Ports

These ports appear in the majority of security alerts and network logs.

Port	Protocol	Service	Primary Use (Legitimate)	Malicious Use (What to Detect)
443	TCP	HTTPS	Encrypted web traffic	Malware C2, encrypted tunnelling, data exfiltration
80	TCP	HTTP	Unencrypted web traffic	Malware delivery, phishing redirects, initial compromise

53	TCP/UDP	DNS	Domain name resolution	DNS tunnelling, DGA-based C2, data exfiltration
445	TCP	SMB	Windows file sharing	Ransomware propagation (EternalBlue), lateral movement
3389	TCP	RDP	Remote desktop access	Brute force attacks, credential stuffing, unauthorised access
22	TCP	SSH	Secure remote administration	Lateral movement, encrypted tunnelling, backdoor access
25 / 587	TCP	SMTP	Email submission and relay	Spam relays, phishing campaigns, BEC attacks

High-Value Ports

These ports are frequently abused and appear in common attack frameworks.

Port	Protocol	Service	Primary Use (Legitimate)	Malicious Use (What to Detect)
135	TCP	RPC	Windows remote procedure calls	Lateral movement (PsExec, WMI), service creation
389 / 636	TCP/UDP	LDAP / LDAPS	Active Directory directory access	AD reconnaissance, privilege escalation enumeration
1433/3306/ 5432	TCP	MSSQL / MySQL / PostgreSQL	Database services	Data exfiltration after SQL injection
20, 21	TCP	FTP	File transfer	Data exfiltration, anonymous file access
123	UDP	NTP	Time synchronisation	NTP amplification DDoS, time-based evasion
161	UDP	SNMP	Network device monitoring	SNMP brute force, device reconnaissance
514	UDP	Syslog	Log forwarding	Log flooding (evasion), log manipulation

Suspicious Ports (Immediate Escalation)

These ports have limited legitimate use in standard enterprise environments. Alerts on these as **destination ports** should be investigated urgently.

Port	Protocol	Service	Why It Is Suspicious
23	TCP	Telnet	Transmits credentials in cleartext. Only acceptable on legacy industrial networks.
69	UDP	TFTP	No authentication. Used for router config theft, PXE boot attacks, and file transfer malware.
4444	TCP	N/A	Default Metasploit reverse shell port. Almost always malicious.
6667	TCP	IRC	Internet Relay Chat. Common for botnet C2 communication, especially in IoT malware.

5555	TCP	N/A	Android ADB (if misconfigured) but more commonly IoT backdoors (Gafgyt, Mirai variants).
1337, 31337	TCP	N/A	"Leet" and "Elite" ports. Common for backdoors, cracked tools, and proof-of-concept malware.
8888	TCP	HTTP-Alt	Alternative proxy and web port. Often used to bypass content filters or host malicious web services.

SOC Alerting Rules

Condition	Recommended Action	Priority
Inbound destination port 3389 from the internet	Verify VPN or jump host requirement. If none, escalate.	High
Outbound destination port 445 from a workstation	Investigate immediately. Potential ransomware lateral movement.	Critical
Destination port 53 with long, high-entropy subdomains	Extract domain names. Check for DNS tunnelling signatures.	High
Any traffic to destination port 4444, 6667, or 5555	Treat as malicious until proven otherwise.	Critical
Outbound destination port 25 from a non-mail server	Investigate for spam relay or compromised host.	High

9. DNS (Domain Name System)

What is DNS

DNS translates human-readable domain names (google.com) into machine-readable IP addresses (142.250.190.46). It is often called the "phonebook of the internet."

DNS Record Types

Record Type	Purpose	Example
A	IPv4 address	google.com → 142.250.190.46
AAAA	IPv6 address	google.com → 2001:4860:4860::8888
CNAME	Canonical name (alias)	www.google.com → google.com
MX	Mail exchange	Email server address
TXT	Text information	SPF, DKIM, DMARC (email security)
NS	Name server	Authoritative DNS servers
PTR	Pointer (reverse DNS)	142.250.190.46 → google.com
SOA	Start of authority	Zone metadata
SRV	Service locator	Service discovery

DNS Resolution Process

Step	Description
1	User types google.com into browser
2	OS checks local DNS cache
3	If not cached, query sent to recursive resolver (ISP)
4	Recursive resolver queries root nameserver
5	Root directs to TLD nameserver (.com)
6	TLD directs to authoritative nameserver for google.com
7	Authoritative nameserver returns A record
8	Recursive resolver returns IP to client
9	Browser connects to IP address

DNS Tunnelling Detection

Indicator	What to Look For
Long subdomains	8f7a3b2c1d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a.example.com
High entropy	Random-looking strings (not dictionary words)
High volume	Thousands of DNS queries from one host
TXT record abuse	Large TXT responses containing encoded data

DNS Attack Types

Attack	Description
DNS Tunneling	Exfiltrating data via DNS queries
DNS Spoofing	Poisoning cache to redirect traffic
DNS Amplification	DDoS using large DNS responses
Domain Generation Algorithms (DGA)	Malware generating random domains for C2

10. DHCP (Dynamic Host Configuration Protocol)

What is DHCP

DHCP automatically assigns IP addresses and network configuration to devices on a network.

DORA Process (4 Steps)

Step	Name	Direction	Description
1	**D**iscover	Client → Server	Client broadcasts "I need an IP"
2	**O**ffer	Server → Client	Server offers an IP address
3	**R**equest	Client → Server	Client accepts the offer

4	**A**cknowledgement	Server → Client	Server confirms assignment

DHCP Message Types

Message	Direction	Purpose
DHCPDISCOVER	Client → Server	Find DHCP servers
DHCPOFFER	Server → Client	Offer IP address
DHCPREQUEST	Client → Server	Request offered IP
DHCPACK	Server → Client	Acknowledge and confirm
DHCPNAK	Server → Client	Decline request
DHCPRELEASE	Client → Server	Release IP address
DHCPINFORM	Client → Server	Request configuration only

DHCP Options

Option	Description
1	Subnet mask
3	Default gateway (router)
6	DNS servers
15	Domain name
42	NTP servers
51	Lease time
53	Message type
54	Server identifier
55	Parameter request list

DHCP Attack Types

Attack	Description	Detection
DHCP Starvation	Attacker requests all available IPs	Many DISCOVER messages from same MAC
Rogue DHCP Server	Attacker sets up fake DHCP server	Multiple OFFERs for same DISCOVER
DHCP Spoofing	Attacker responds as legitimate server	Unexpected gateway/DNS assignments

11. ARP (Address Resolution Protocol)

What is ARP

ARP maps IP addresses to MAC addresses within a local network.

ARP Request and Reply

Step	Direction	Description
1	Host A → Broadcast	"Who has IP 192.168.1.1?"
2	Host B → Host A	"I do. My MAC is AA:BB:CC:DD:EE:FF"

ARP Packet Structure

Field	Size	Description
Hardware Type	16 bits	Ethernet (1)
Protocol Type	16 bits	IPv4 (0x0800)
Hardware Length	8 bits	MAC address length (6)
Protocol Length	8 bits	IP address length (4)
Operation	16 bits	1 = Request, 2 = Reply
Sender MAC	48 bits	Source MAC address
Sender IP	32 bits	Source IP address
Target MAC	48 bits	Target MAC (zero for request)
Target IP	32 bits	Target IP address

ARP Cache

Attribute	Description
Purpose	Stores IP-to-MAC mappings to avoid repeated lookups
Lifetime	Typically 15-60 seconds (short to prevent poisoning)
View cache	arp -a (Windows/Linux)

ARP Poisoning (ARP Spoofing)

Attack	Description
What it is	Attacker sends fake ARP replies to associate their MAC with a legitimate IP
Effect	Traffic intended for legitimate host is sent to attacker
Use	Man-in-the-middle attacks, session hijacking
Detection	Multiple MAC addresses claiming same IP

Detection Commands

Command	What to Look For
arp -a	Duplicate IPs with different MACs

Wireshark filter arp	Multiple gratuitous ARP replies
ip neigh show (Linux)	Suspicious ARP entries
tshark -r capture.pcap -Y "arp.opcode == 2 and arp.src.proto_ipv4 == arp.dst.proto_ipv4"	Look for unexpected gratuitous ARPs

12. ICMP (Internet Control Message Protocol)

What is ICMP

ICMP is used for error reporting and diagnostic functions (ping, traceroute). It is not used for user data transmission.

Common ICMP Types

Type	Code	Name	Description
0	-	Echo Reply	Response to ping
3	0	Net Unreachable	Network unreachable
3	1	Host Unreachable	Host unreachable
3	3	Port Unreachable	Port unreachable (UDP)
3	4	Fragmentation Needed	DF set but fragmentation required (PMTUD)
5	-	Redirect	Better route available
8	-	Echo Request	Ping request
11	0	Time Exceeded (TTL expired)	TTL reached zero (traceroute)
11	1	Time Exceeded (Reassembly timeout)	Fragment reassembly timeout

ICMPv6 Types (for IPv6)

Type	Name	Description
128	Echo Request	Ping request
129	Echo Reply	Ping response
133	Router Solicitation	Request router information
134	Router Advertisement	Router information
135	Neighbor Solicitation	ARP equivalent
136	Neighbor Advertisement	ARP reply equivalent

ICMP Attacks

Attack	Description
ICMP Flood (Ping Flood)	DDoS using thousands of ping packets
Smurf Attack	Amplified ping to broadcast address
Ping of Death	Oversized packet causes buffer overflow

ICMP Tunneling	Data exfiltration inside ICMP packets

Detection Relevance

Use Case	ICMP Role
Ping sweeps	Host discovery (Type 8 requests)
Network mapping	Traceroute (Type 11)
Firewall testing	Destination unreachable (Type 3)

13. HTTP and HTTPS

HTTP (Hypertext Transfer Protocol)

Attribute	Description
Port	80
Encryption	None (plain text)
Security	Not secure — all data visible
Use	Legacy web traffic

Modern HTTP Versions:

- **HTTP/2:** Binary protocol, multiplexed. Widely used. Many WAFs inspect it differently than HTTP/1.1.
- **HTTP/3:** Uses QUIC over UDP 443, not TCP. Many security tools cannot inspect it. Alert on UDP 443 as potentially suspicious.

HTTP Request Methods

Method	Purpose	Security Relevance
GET	Retrieve data	Should not change server state
POST	Submit data	Used for forms, logins
PUT	Upload data	File uploads
DELETE	Remove data	Dangerous if exposed
HEAD	Retrieve headers only	Reconnaissance
PATCH	Partial update	Modification
OPTIONS	Query capabilities	Reconnaissance

HTTP Status Codes

Code Range	Category	Example
1xx	Informational	100 Continue
2xx	Success	200 OK, 201 Created
3xx	Redirection	301 Moved, 302 Found

4xx	Client Error	401 Unauthorized, 403 Forbidden, 404 Not Found
5xx	Server Error	500 Internal Error, 503 Unavailable

HTTP Headers Important for Security

Header	Purpose
User-Agent	Identifies client software (can reveal scanners)
Referer	Source of the request
Cookie	Session tokens (session hijacking target)
Authorization	Credentials (should only be over HTTPS)
X-Forwarded-For	Original client IP behind proxy
Content-Type	Data format (MIME type)
Content-Length	Size of request body

HTTPS (HTTP Secure)

Attribute	Description
Port	443
Encryption	TLS (Transport Layer Security)
Authentication	Server certificates (optional client certs)
Security	Confidentiality, integrity, authentication

TLS Versions

Version	Status	Security
SSLv2, SSLv3	Deprecated	Insecure
TLS 1.0	Deprecated	Weak
TLS 1.1	Deprecated	Weak
TLS 1.2	Current	Secure
TLS 1.3	Current	Most secure

TLS Handshake (Simplified)

Step	Description
1	Client sends "Client Hello" (supported cipher suites)
2	Server sends "Server Hello" (chosen cipher suite) + certificate
3	Client verifies certificate, generates session key
4	Encrypted communication begins

14. Firewalls and Network Security

Types of Firewalls

Type	Layer	How It Works	Example
Packet Filtering	Layer 3-4	Filters based on IP, port, protocol	iptables, ACLs
Stateful Inspection	Layer 4	Tracks connection state	Check Point, pfSense
Application Gateway	Layer 7	Inspects application data	Web Application Firewall (WAF)
Next-Generation (NGFW)	Layers 3-7	Application ID, user ID, IPS	Palo Alto, Fortinet

Firewall Rules

Component	Description
Source	Where traffic comes from (IP, subnet, interface)
Destination	Where traffic is going (IP, subnet, interface)
Port/Protocol	Service being accessed (TCP 443, UDP 53)
Action	Allow, Deny, Reject, Log
Direction	Inbound, Outbound, Forward

Firewall Rule Example

```
1. allow inbound from any to 10.0.0.5 port 443 (HTTPS)
2. deny inbound from any to any port 3389 (RDP)
3. allow outbound from 10.0.0.0/8 to any port 443
4. deny outbound from any to any port 25 (SMTP)
```

Network Segmentation

Concept	Description	Security Benefit
DMZ	Public-facing servers (web, email)	Isolates external-facing systems
VLAN	Virtual LAN segmentation	Contains lateral movement
Air Gap	Physically disconnected network	Highest security
Zero Trust	Never trust, always verify	Eliminates implicit trust

IDS vs IPS

Type	Placement	Action	Purpose
IDS	Monitor-only	Alert	Detect and log
IPS	In-line	Alert + Block	Detect and prevent

15. NAT (Network Address Translation)

What is NAT

NAT translates private IP addresses to public IP addresses, allowing multiple devices to share a single public IP.

Types of NAT

Type	Description	Example
Static NAT	One-to-one mapping	10.0.0.5 → 203.0.113.10
Dynamic NAT	Pool of public IPs	First available from pool
PAT (Port Address Translation)	Many-to-one using ports	10.0.0.5:12345 → 203.0.113.1:54321

How PAT Works (Most Common)

Step	Description
1	Device on private network sends packet with source port
2	Router replaces source IP with public IP, assigns unique source port
3	Router tracks mapping in NAT table
4	Return traffic is mapped back to original device

NAT Table Example

Private IP:Port	Public IP:Port	Destination
10.0.0.5:12345	203.0.113.1:54321	8.8.8.8:443
10.0.0.6:54321	203.0.113.1:12345	1.1.1.1:53

16. VLANs and Network Segmentation

What is a VLAN

A VLAN (Virtual Local Area Network) is a logical network segment that groups devices together regardless of physical location.

Benefits of VLANs

Benefit	Description
Security	Isolates sensitive systems
Performance	Reduces broadcast traffic
Flexibility	Devices can move without reconfiguration
Cost	Fewer physical switches needed

Common VLANs in Enterprise

VLAN ID	Purpose	Security Level
1	Native/default	Often compromised, avoid using

10	Management	Restricted access
20	User devices	Standard
30	Servers	Higher security
40	DMZ	Public-facing, isolated
50	Guest	Internet only
99	Blackhole	Unused ports

VLAN Tagging (802.1Q)

Field	Size	Description
TPID	16 bits	Tag protocol ID (0x8100)
Priority	3 bits	QoS
CFI	1 bit	Canonical format indicator
VID	12 bits	VLAN ID (1-4094)

VLAN Hopping Attacks

Attack	Description
Switch Spoofing	Attacker pretends to be a switch
Double Tagging	Two VLAN tags to bypass segmentation

17. VPNs (Virtual Private Networks)

What is a VPN

A VPN creates an encrypted tunnel between two endpoints over an untrusted network (usually the internet).

VPN Protocols

Protocol	Encryption	Use Case
IPsec	Strong	Site-to-site, remote access
SSL/TLS	Strong	Remote access (web-based)
OpenVPN	Strong	Remote access
WireGuard	Strong	Modern, fast
PPTP	Weak	Deprecated (insecure)
L2TP/IPsec	Moderate	Legacy

VPN Use Cases

Use Case	Description
Remote Access	Employee connects from home

Site-to-Site	Office networks connected
Cloud Access	Secure connection to cloud provider
Privacy	Hide traffic from ISP

VPN Detection

Indicator	What It Looks Like
ESP packets (IP protocol 50)	IPsec traffic
UDP port 500 or 4500	IKE (IPsec key exchange)
UDP port 1194	OpenVPN
UDP port 51820	WireGuard
TCP port 443 (non-HTTP patterns)	SSL VPN

18. Load Balancers and Proxies

Load Balancer

Attribute	Description
Purpose	Distributes traffic across multiple servers
Types	Layer 4 (transport), Layer 7 (application)
Algorithms	Round-robin, least connections, IP hash
Health checks	Verifies servers are responding

Load Balancer Deployment

Type	Description
Inline	Traffic passes through load balancer
One-arm	Load balancer on same subnet as servers

Proxy Types

Type	Direction	Purpose
Forward Proxy	Client → Internet	Access control, caching, anonymity
Reverse Proxy	Internet → Server	Load balancing, security, SSL termination
Transparent Proxy	Intercepts without configuration	Content filtering
SOCKS Proxy	Generic proxy	Any protocol

Proxy Detection

Indicator	What It Looks Like
X-Forwarded-For header	Original client IP behind proxy

Via header	Proxy server identification
X-Cache header	Cache hit/miss

19. Network Troubleshooting Commands

Windows Commands

Command	Purpose	Example
ipconfig	View IP configuration	ipconfig /all
ping	Test connectivity	ping 8.8.8.8
tracert	Trace route	tracert google.com
nslookup	DNS lookup	nslookup google.com
netstat	Connection statistics	netstat -an
arp	View ARP cache	arp -a
route	View routing table	route print
netsh	Network configuration	netsh advfirewall show allprofiles
nmap (install)	Port scanning	nmap -sS 192.168.1.1

Linux Commands

Command	Purpose	Example
ip addr	View IP configuration	ip addr show
ping	Test connectivity	ping -c 4 8.8.8.8
traceroute	Trace route	traceroute google.com
dig	DNS lookup	dig google.com
nslookup	DNS lookup	nslookup google.com
ss	Socket statistics	ss -tulpn
netstat	Network statistics	netstat -tulpn
arp	View ARP cache	arp -n
ip route	Routing table	ip route show
tcpdump	Packet capture	tcpdump -i eth0

Key Flags Reference

Command	Flag	Meaning
ping	-c	Count (Linux)
ping	-n	Count (Windows)

ping	-t	Continuous (Windows)
tracert/traceroute	-d	Do not resolve hostnames
netstat	-a	All connections
netstat	-n	Numeric (no hostname resolution)
netstat	-o	Owning process ID (Windows)
netstat	-p	Protocol (Linux)
ss	-t	TCP sockets
ss	-u	UDP sockets
ss	-l	Listening sockets
ss	-p	Process information
ss	-n	Numeric

20. Packet Capture and Analysis

Capture Commands

Tool	Command	Purpose
tcpdump	tcpdump -i eth0 -c 100	Capture 100 packets
tcpdump	tcpdump -i eth0 -w capture.pcap	Write to file
tcpdump	tcpdump -r capture.pcap	Read capture file
tshark	tshark -i eth0 -c 100	Capture with Wireshark tools

Wireshark Display Filters

Filter	Purpose
ip.addr == 192.168.1.1	Traffic to/from IP
tcp.port == 80	HTTP traffic
udp.port == 53	DNS traffic
http.request	HTTP requests only
dns.qry.name contains "google"	DNS queries for google
tcp.flags.syn == 1 and tcp.flags.ack == 0	SYN scans
tcp.analysis.ack_rtt > 1	Slow responses

Capture Filters (tcpdump/tshark)

Filter	Purpose
host 192.168.1.1	Traffic to/from IP

port 80	Traffic on port 80
src host 192.168.1.1	Traffic from IP
dst host 192.168.1.1	Traffic to IP
tcp and port 22	TCP port 22 only
not arp	Exclude ARP

PCAP Analysis Questions

Question	How to Answer
What IPs communicated?	tshark -r file.pcap -T fields -e ip.src -e ip.dst | sort | uniq
What protocols were used?	tshark -r file.pcap -T fields -e frame.protocols | sort | uniq -c
What was the top talker?	tshark -r file.pcap -T fields -e ip.src | sort | uniq -c | sort -nr
What HTTP requests were made?	tshark -r file.pcap -Y http.request -T fields -e http.request.uri

21. Quick Reference Tables (Keywords and Definitions)

A

ACK	Acknowledgment flag in TCP, confirms receipt of data
A Record	DNS record mapping domain name to IPv4 address
AAAA Record	DNS record mapping domain name to IPv6 address
ARP	Address Resolution Protocol, maps IP to MAC addresses
Application Layer	Layer 7 of OSI, where user applications operate

B

Bandwidth	Maximum data transfer rate of a network
BGP	Border Gateway Protocol, core routing protocol of the internet
Broadcast	Sending data to all devices on a network
Broadcast Address	IP address that reaches all hosts on a subnet

C

CIDR	Classless Inter-Domain Routing, modern IP addressing scheme
CNAME	Canonical Name record, domain alias in DNS
Collision Domain	Network segment where packet collisions can occur
CRC	Cyclic Redundancy Check, error detection code

D

DDoS	Distributed Denial of Service, attack from multiple sources
Default Gateway	Router that forwards traffic outside local network
DHCP	Dynamic Host Configuration Protocol, assigns IP addresses automatically
DNS	Domain Name System, translates domain names to IP addresses
DMZ	Demilitarized Zone, network segment for public-facing servers
DupACK	Duplicate acknowledgment in TCP

E

Ethernet	Most common LAN technology, IEEE 802.3
EtherType	Field in Ethernet frame indicating protocol type

F

FIN	Finish flag in TCP, graceful connection termination
Firewall	Device that filters traffic based on rules
Fragment	Divided piece of an IP packet

Frame	Layer 2 data unit (Ethernet)

G

Gateway	Device that connects different networks
Gratuitous ARP	ARP reply sent without a request (often malicious)

H

Handshake	Initial connection setup process (TCP three-way handshake)
Hop	One segment of a path between two routers
HTTP	Hypertext Transfer Protocol, web traffic
HTTPS	HTTP over TLS/SSL, encrypted web traffic

I

IaaS	Infrastructure as a Service, cloud computing model
ICMP	Internet Control Message Protocol, used for ping and traceroute
IDS	Intrusion Detection System, monitors and alerts
IKE	Internet Key Exchange, used in IPsec VPNs
IP	Internet Protocol, handles addressing and routing
IPsec	IP Security, VPN protocol suite
IPv4	Internet Protocol version 4, 32-bit addresses
IPv6	Internet Protocol version 6, 128-bit addresses
IPS	Intrusion Prevention System, blocks malicious traffic

J

Jitter	Variation in packet arrival time
Jumbo Frame	Ethernet frame larger than 1500 bytes

K

Kernel	Operating system core, includes network stack

L

LAN	Local Area Network, network in limited geographic area
Latency	Delay between sending and receiving data
Layer 2	Data Link layer, MAC addresses and switching
Layer 3	Network layer, IP addresses and routing
Layer 4	Transport layer, TCP and UDP ports
Layer 7	Application layer, HTTP, DNS, SMTP

Load Balancer	Device that distributes traffic across servers
Loopback	Virtual interface for local communication (127.0.0.1, ::1)

M

MAC Address	Media Access Control address, hardware identifier
MSS	Maximum Segment Size, TCP parameter
MTU	Maximum Transmission Unit, largest packet size
Multicast	Sending data to a group of interested devices
MX Record	Mail Exchange record, directs email servers

N

NAT	Network Address Translation, maps private to public IPs
NetBIOS	Legacy Windows naming service
Netflow	Network flow monitoring protocol
NIC	Network Interface Card
NS Record	Name Server record, identifies DNS servers

O

OSI Model	Open Systems Interconnection model, 7-layer framework
OUI	Organizationally Unique Identifier, first 3 bytes of MAC

P

Packet	Layer 3 data unit (IP packet)
PaaS	Platform as a Service, cloud computing model
PAT	Port Address Translation, type of NAT
PCAP	Packet capture file format
Port	Transport layer endpoint (0-65535)
Protocol	Set of rules for communication
Proxy	Intermediate device that forwards requests
PSH	Push flag in TCP, send data immediately
PTR Record	Pointer record, reverse DNS mapping

Q

QoS	Quality of Service, traffic prioritisation
Query	Request in DNS or other client-server protocols

R

RARP	Reverse ARP, obsolete
RDP	Remote Desktop Protocol, port 3389
RST	Reset flag in TCP, abrupt connection termination
Router	Device that forwards packets between networks
RTT	Round Trip Time, time for packet to go and return
RTO	Retransmission Timeout, TCP parameter

S

aaS	Software as a Service, cloud computing model
Segment	Layer 4 data unit (TCP segment)
SIEM	Security Information and Event Management
SMB	Server Message Block, Windows file sharing, port 445
SMTP	Simple Mail Transfer Protocol, email sending
SNMP	Simple Network Management Protocol
Socket	IP address + port combination
SPF	Sender Policy Framework, email authentication
SSH	Secure Shell, remote access, port 22
Subnet	Logical subdivision of an IP network
Switch	Device that forwards frames within a network
SYN	Synchronise flag in TCP, initiates connection

T

TCP	Transmission Control Protocol, reliable, connection-oriented
TCP/IP	Suite of protocols for internet communication
TLS	Transport Layer Security, encrypts network traffic
TTL	Time To Live, maximum hops before packet is discarded
TXT Record	Text record in DNS, used for SPF, DKIM, DMARC
Three-way Handshake	SYN, SYN-ACK, ACK process to establish TCP connection

U

UDP	User Datagram Protocol, unreliable, connectionless
Unicast	One-to-one communication
URG	Urgent flag in TCP

V

VLAN	Virtual LAN, logical network segmentation
VPN	Virtual Private Network, encrypted tunnel

W

WAN	Wide Area Network, network spanning large geographic area
Wireshark	Network protocol analyzer
WPA/WPA2/WPA3	Wi-Fi Protected Access, wireless security protocols

Z

Zero-Day	**A vulnerability unknown to vendors with no available patch**
Zero Trust	Security model requiring continuous verification
Zombie	Compromised computer used in DDoS attacks